

Mathematical Foundations of Reinforcement Learning

Homework 1 – Solutions

Question 1

(a)

For a state $x \in \mathcal{X}$, define its period as

$$d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\}.$$

Since the Markov chain is irreducible, all states have the same period.

Suppose that some state x has a self-loop, i.e.,

$$P(x, x) > 0.$$

Then

$$P^1(x, x) > 0,$$

so 1 belongs to the set of possible return times to x . Therefore,

$$d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\} = 1.$$

Since the chain is irreducible, every state has the same period. Therefore every state has period 1, and the Markov chain is aperiodic.

(b)

Consider the Markov chain with state space

$$\mathcal{X} = \{1, 2, 3\}$$

and transition matrix

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

There are no self-loops because

$$P(i, i) = 0, \quad i = 1, 2, 3.$$

The chain is irreducible because every state can reach every other state in one step.

For state 1, we can return in two steps:

$$1 \rightarrow 2 \rightarrow 1,$$

and in three steps:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1.$$

Thus

$$P^2(1, 1) > 0 \quad \text{and} \quad P^3(1, 1) > 0.$$

Hence

$$d(1) = \gcd(2, 3, \dots) = 1.$$

By irreducibility, all states have period 1. Therefore the chain is irreducible and aperiodic, while having no self-loops.

Question 2

Let P be an irreducible transition matrix with period b .

(a)

Fix a state $x_0 \in \mathcal{X}$. For $i = 1, \dots, b$, define

$$C_i = \{y \in \mathcal{X} : \text{there exists a path from } x_0 \text{ to } y \text{ of length } i - 1 \pmod{b}\}.$$

We first show that this definition is well-defined.

Suppose there are two paths from x_0 to y having lengths m and n . Since the chain is irreducible, there exists a path from y back to x_0 , say of length r . Therefore,

$$m + r \quad \text{and} \quad n + r$$

are both return times to x_0 .

Since the period is b , both return times are divisible by b :

$$m + r \equiv 0 \pmod{b}, \quad n + r \equiv 0 \pmod{b}.$$

Subtracting gives

$$m \equiv n \pmod{b}.$$

Thus every state belongs to exactly one of the sets C_i . Irreducibility implies that every state is reachable from x_0 , so

$$\mathcal{X} = C_1 \cup C_2 \cup \dots \cup C_b,$$

where the sets are pairwise disjoint.

Now suppose

$$x \in C_i \quad \text{and} \quad P(x, y) > 0.$$

A path from x_0 to x has length congruent to $i - 1$ modulo b . Adding the transition from x to y gives a path to y whose length is congruent to i modulo b .

Therefore,

$$y \in C_{i+1},$$

where the index is interpreted modulo b , so that

$$C_{b+1} = C_1.$$

Hence

$$P(x, y) > 0$$

only if

$$x \in C_i \quad \text{and} \quad y \in C_{i+1}$$

for some i .

(b)

Let

$$n_i = |C_i|, \quad N_i = \sum_{j=1}^i n_j.$$

If the states are ordered according to

$$C_1, C_2, \dots, C_b,$$

then the transition matrix has the block structure

$$P = \begin{pmatrix} 0 & P_{12} & 0 & \cdots & 0 \\ 0 & 0 & P_{23} & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & P_{b-1,b} \\ P_{b1} & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

The block $P_{i,i+1}$ has dimension

$$n_i \times n_{i+1},$$

while P_{b1} has dimension

$$n_b \times n_1.$$

Thus the only potentially nonzero blocks correspond to transitions from C_i to C_{i+1} , including the transition from C_b back to C_1 .

Question 3

(a)

Let

$$|\mathcal{X}| = n.$$

The uniform distribution is

$$\pi(x) = \frac{1}{n}, \quad x \in \mathcal{X}.$$

For any $y \in \mathcal{X}$,

$$(\pi^T P)(y) = \sum_{x \in \mathcal{X}} \pi(x) P(x, y) = \frac{1}{n} \sum_{x \in \mathcal{X}} P(x, y).$$

Since P is symmetric,

$$P(x, y) = P(y, x).$$

Therefore,

$$\sum_{x \in \mathcal{X}} P(x, y) = \sum_{x \in \mathcal{X}} P(y, x) = 1.$$

Consequently,

$$(\pi^T P)(y) = \frac{1}{n} = \pi(y).$$

Thus

$$\boxed{\pi^T P = \pi^T},$$

so the uniform distribution is stationary.

(b)

Yes. Symmetry is sufficient but not necessary.

Consider

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

This matrix is not symmetric. However, every row and every column sums to one, so P is doubly stochastic.

Let

$$\pi = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}.$$

Then

$$\pi^T P = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} P = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \pi^T.$$

Therefore the uniform distribution is stationary even though P is not symmetric.

Question 4

(a)

The average reward value function is

$$V_T(x_0) = \frac{1}{T+1} \mathbb{E} \left[\sum_{t=0}^T R(X_t) \middle| X_0 = x_0 \right].$$

By linearity of expectation,

$$V_T(x_0) = \frac{1}{T+1} \sum_{t=0}^T \mathbb{E}[R(X_t) \mid X_0 = x_0].$$

Moreover,

$$\mathbb{E}[R(X_t) \mid X_0 = x_0] = \sum_{x \in \mathcal{X}} P^t(x_0, x) R(x).$$

Since the Markov chain is irreducible and aperiodic,

$$P^t(x_0, x) \rightarrow \pi(x)$$

as $t \rightarrow \infty$, where π is the unique stationary distribution.

Since R is bounded,

$$\mathbb{E}[R(X_t) \mid X_0 = x_0] \rightarrow \sum_{x \in \mathcal{X}} \pi(x) R(x).$$

The Cesaro average of a convergent sequence converges to the same limit. Therefore,

$$\boxed{V(x_0) = \lim_{T \rightarrow \infty} V_T(x_0) = \sum_{x \in \mathcal{X}} \pi(x) R(x)}.$$

(b)

Define

$$Y_n = X_{n, \tau_{\text{mix}}(\epsilon)}$$

and

$$Z_n = R(Y_n).$$

Then

$$\hat{V}_N = \frac{1}{N} \sum_{n=1}^N Z_n.$$

The simulations are independent, so the Z_n are IID.

Using the bias-variance decomposition,

$$\mathbb{E}[(\hat{V}_N - V)^2] = \text{Var}(\hat{V}_N) + (\mathbb{E}[\hat{V}_N] - V)^2.$$

Since

$$|Z_n| \leq R_{\max},$$

Popoviciu's inequality gives

$$\text{Var}(Z_n) \leq \frac{(2R_{\max})^2}{4} = R_{\max}^2.$$

Hence

$$\text{Var}(\hat{V}_N) = \frac{\text{Var}(Z_n)}{N} \leq \frac{R_{\max}^2}{N}.$$

By the definition of mixing time,

$$\left\| P^{\tau_{\text{mix}}(\epsilon)}(x_0, \cdot) - \pi \right\|_{\text{TV}} \leq \epsilon.$$

Therefore,

$$\begin{aligned} |\mathbb{E}[\hat{V}_N] - V| &= \left| \sum_{x \in \mathcal{X}} \left(P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x) \right) R(x) \right| \\ &\leq R_{\max} \sum_{x \in \mathcal{X}} \left| P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x) \right| \\ &\leq 2R_{\max}\epsilon. \end{aligned}$$

Thus

$$(\mathbb{E}[\hat{V}_N] - V)^2 \leq 4R_{\max}^2\epsilon^2.$$

Combining the two bounds,

$$\boxed{\mathbb{E}[(\hat{V}_N - V)^2] \leq R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right).}$$

(c)

Let $\epsilon > 0$ denote the desired MSE accuracy. Introduce a separate mixing accuracy η . From part (b),

$$\mathbb{E}[(\hat{V}_N - V)^2] \leq R_{\max}^2 \left(\frac{1}{N} + 4\eta^2 \right).$$

Choose

$$N = \left\lceil \frac{2R_{\max}^2}{\epsilon} \right\rceil$$

and

$$\eta = \sqrt{\frac{\epsilon}{8R_{\max}^2}}.$$

Then

$$\frac{R_{\max}^2}{N} \leq \frac{\epsilon}{2}$$

and

$$4R_{\max}^2\eta^2 = \frac{\epsilon}{2}.$$

Hence

$$\mathbb{E}[(\hat{V}_N - V)^2] \leq \epsilon.$$

Using

$$\tau_{\text{mix}}(\eta) \leq \lceil \log_2(\eta^{-1}) \rceil \tau_{\text{mix}}(1/4),$$

we have

$$\log_2(\eta^{-1}) = \frac{1}{2} \log_2 \left(\frac{8R_{\text{max}}^2}{\epsilon} \right).$$

Therefore,

$$\tau_{\text{mix}}(\eta) \leq \left\lceil \frac{1}{2} \log_2 \left(\frac{8R_{\text{max}}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}}(1/4).$$

Thus the total number of Markov-chain samples is bounded by

$$\left\lceil \frac{2R_{\text{max}}^2}{\epsilon} \right\rceil \left\lceil \frac{1}{2} \log_2 \left(\frac{8R_{\text{max}}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}}(1/4).$$

Question 5

Suppose $T : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction:

$$\|T(x) - T(y)\| \leq \alpha \|x - y\|, \quad 0 \leq \alpha < 1.$$

(a)

Suppose x^* and y^* are both fixed points. Then

$$T(x^*) = x^*, \quad T(y^*) = y^*.$$

Therefore,

$$\begin{aligned} \|x^* - y^*\| &= \|T(x^*) - T(y^*)\| \\ &\leq \alpha \|x^* - y^*\|. \end{aligned}$$

Since $\alpha < 1$, this is possible only if

$$\|x^* - y^*\| = 0.$$

Hence

$$x^* = y^*.$$

Thus the fixed point is unique.

(b)

Consider

$$\mathcal{X} = (0, 1)$$

and

$$T(x) = \frac{x}{2}.$$

For any $x, y \in \mathcal{X}$,

$$|T(x) - T(y)| = \frac{1}{2} |x - y|.$$

Thus T is a contraction with $\alpha = 1/2$.

A fixed point would satisfy

$$x = \frac{x}{2},$$

which gives

$$x = 0.$$

However,

$$0 \notin (0, 1).$$

Therefore T has no fixed point in $\mathcal{X}$.

(c)

Choose any $x_0 \in \mathcal{X}$ and define

$$x_{k+1} = T(x_k).$$

By contractivity,

$$\|x_{k+1} - x_k\| \leq \alpha \|x_k - x_{k-1}\|.$$

Repeated application gives

$$\|x_{k+1} - x_k\| \leq \alpha^k \|x_1 - x_0\|.$$

For $m > n$,

$$\begin{aligned} \|x_m - x_n\| &\leq \sum_{k=n}^{m-1} \|x_{k+1} - x_k\| \\ &\leq \|x_1 - x_0\| \sum_{k=n}^{m-1} \alpha^k \\ &\leq \frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\|. \end{aligned}$$

Since $\alpha < 1$, the right-hand side converges to zero as $n \rightarrow \infty$. Hence (x_k) is a Cauchy sequence.

Since $\mathcal{X}$ is complete, there exists $x^* \in \mathcal{X}$ such that

$$x_k \rightarrow x^*.$$

Since a contraction is continuous,

$$T(x^*) = T\left(\lim_{k \rightarrow \infty} x_k\right) = \lim_{k \rightarrow \infty} T(x_k) = \lim_{k \rightarrow \infty} x_{k+1} = x^*.$$

Thus x^* is a fixed point. By part (a), it is unique.

Question 6

The algorithm is

$$x(k+1) = Wx(k),$$

where

$$W\mathbf{1} = \mathbf{1} \quad \text{and} \quad \mathbf{1}^T W = \mathbf{1}^T.$$

(a)

Suppose W were contractive in some norm. Then there would exist $\alpha < 1$ such that

$$\|Wx - Wy\| \leq \alpha \|x - y\|$$

for all x, y .

Choose

$$x = \mathbf{1}, \quad y = 0.$$

Since

$$W\mathbf{1} = \mathbf{1},$$

we obtain

$$\|W\mathbf{1} - W0\| = \|\mathbf{1}\|.$$

Contractivity would imply

$$\|\mathbf{1}\| \leq \alpha \|\mathbf{1}\|,$$

which is impossible for $\alpha < 1$.

Therefore W is not contractive in any norm.

(b)

A fixed point satisfies

$$Wx = x.$$

Thus the fixed points are exactly the eigenvectors associated with the eigenvalue 1.

Since

$$W\mathbf{1} = \mathbf{1},$$

$\mathbf{1}$ is an eigenvector corresponding to eigenvalue 1.

The matrix W is irreducible and nonnegative. By the Perron–Frobenius theorem, the eigenspace associated with the spectral radius 1 is one-dimensional.

Therefore,

$$\boxed{\text{Fix}(W) = \text{Span}(\mathbf{1}) = \{a\mathbf{1} : a \in \mathbb{R}\}.$$

(c)

Let

$$\mathcal{A}(\bar{x}) = \left\{ x \in \mathbb{R}^N : \frac{1}{N} \mathbf{1}^T x = \bar{x} \right\}.$$

If $x \in \mathcal{A}(\bar{x})$, then

$$\frac{1}{N} \mathbf{1}^T x = \bar{x}.$$

Using

$$\mathbf{1}^T W = \mathbf{1}^T,$$

we obtain

$$\frac{1}{N} \mathbf{1}^T Wx = \frac{1}{N} \mathbf{1}^T x = \bar{x}.$$

Therefore,

$$Wx \in \mathcal{A}(\bar{x}).$$

(d)

Define

$$J = \frac{1}{N} \mathbf{1} \mathbf{1}^T.$$

For $x \in \mathcal{A}(\bar{x})$,

$$Jx = \frac{1}{N} \mathbf{1} \mathbf{1}^T x = \bar{x} \mathbf{1}.$$

Also,

$$W \bar{x} \mathbf{1} = \bar{x} \mathbf{1}.$$

Hence

$$\begin{aligned} Wx - \bar{x} \mathbf{1} &= W(x - \bar{x} \mathbf{1}) \\ &= (W - J)(x - \bar{x} \mathbf{1}), \end{aligned}$$

because

$$J(x - \bar{x} \mathbf{1}) = 0.$$

Therefore,

$$\|Wx - \bar{x} \mathbf{1}\|_2 \leq \left\| W - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right\|_2 \|x - \bar{x} \mathbf{1}\|_2.$$

Define

$$\sigma = \left\| W - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right\|_2.$$

Under the assumptions of the problem,

$$\sigma < 1.$$

Hence

$$\boxed{\|Wx - \bar{x} \mathbf{1}\|_2 \leq \sigma \|x - \bar{x} \mathbf{1}\|_2.}$$

Thus W is a pseudo-contraction on $\mathcal{A}(\bar{x})$.

(e)

Assume W is symmetric. Then W has an orthonormal eigenbasis $v_1, \dots, v_N$ with eigenvalues

$$1 = \lambda_1, \lambda_2, \dots, \lambda_N,$$

ordered such that

$$1 = |\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_N|.$$

We can choose

$$v_1 = \frac{1}{\sqrt{N}} \mathbf{1}.$$

Write

$$x(0) = \sum_{i=1}^N c_i v_i.$$

Since

$$c_1 v_1 = \bar{x} \mathbf{1},$$

we have

$$x(k) - \bar{x} \mathbf{1} = \sum_{i=2}^N c_i \lambda_i^k v_i.$$

Therefore,

$$\begin{aligned}\|x(k) - \bar{x}\mathbf{1}\|_2^2 &= \sum_{i=2}^N c_i^2 |\lambda_i|^{2k} \\ &\leq |\lambda_2|^{2k} \sum_{i=2}^N c_i^2.\end{aligned}$$

Define

$$A = \left(\sum_{i=2}^N c_i^2 \right)^{1/2}.$$

Then

$$\boxed{\|x(k) - \bar{x}\mathbf{1}\|_2 \leq A |\lambda_2|^k.}$$

Question 7

The iteration is

$$x(k+1) = x(k) - \gamma(\nabla f(x(k)) + e(k)).$$

Since x^* minimizes f ,

$$\nabla f(x^*) = 0.$$

Define

$$d_k = x(k) - x^*.$$

Then

$$d_{k+1} = d_k - \gamma \nabla f(x(k)) - \gamma e(k).$$

Taking squared norms and expectations, the cross term involving $e(k)$ vanishes because $e(k)$ has zero mean and is independent of $x(k)$. Thus,

$$\mathbb{E}\|d_{k+1}\|^2 = \mathbb{E}\|d_k - \gamma \nabla f(x(k))\|^2 + \gamma^2 \sigma^2.$$

Expanding,

$$\|d_k - \gamma \nabla f(x(k))\|^2 = \|d_k\|^2 - 2\gamma \langle \nabla f(x(k)), d_k \rangle + \gamma^2 \|\nabla f(x(k))\|^2.$$

Using the inequality given in the problem with $y = x^*$,

$$\langle \nabla f(x(k)), d_k \rangle \geq \frac{L\mu}{L+\mu} \|d_k\|^2 + \frac{1}{L+\mu} \|\nabla f(x(k))\|^2.$$

Therefore,

$$\begin{aligned}\|d_k - \gamma \nabla f(x(k))\|^2 &\leq \left(1 - \frac{2\gamma L\mu}{L+\mu}\right) \|d_k\|^2 \\ &\quad + \left(\gamma^2 - \frac{2\gamma}{L+\mu}\right) \|\nabla f(x(k))\|^2.\end{aligned}$$

Since

$$0 < \gamma < \frac{2}{L+\mu},$$

we have

$$\gamma^2 - \frac{2\gamma}{L+\mu} < 0.$$

Hence

$$\mathbb{E}\|d_{k+1}\|^2 \leq q\mathbb{E}\|d_k\|^2 + \gamma^2\sigma^2,$$

where

$$q = 1 - \frac{2\gamma L\mu}{L + \mu}.$$

Iterating,

$$\mathbb{E}\|d_{k+1}\|^2 \leq q^{k+1}\mathbb{E}\|d_0\|^2 + \gamma^2\sigma^2 \sum_{i=0}^k q^i.$$

Since

$$\sum_{i=0}^k q^i \leq \frac{1}{1-q}$$

and

$$1 - q = \frac{2\gamma L\mu}{L + \mu},$$

we obtain

$$\boxed{\mathbb{E}\|x(k+1) - x^*\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)^{k+1} \mathbb{E}\|x(0) - x^*\|^2 + \frac{\gamma\sigma^2(\mu + L)}{2\mu L}.$$

As $k \rightarrow \infty$,

$$\limsup_{k \rightarrow \infty} \mathbb{E}\|x(k) - x^*\|^2 \leq \frac{\gamma\sigma^2(\mu + L)}{2\mu L}.$$

Thus, by choosing γ sufficiently small, the asymptotic error can be made arbitrarily small. However, a smaller γ makes

$$1 - \frac{2\gamma\mu L}{\mu + L}$$

closer to 1, which slows down convergence.

Therefore, the cost of higher solution accuracy is a larger number of iterations.

Question 8

We construct a sequence that converges in probability to 0 but does not converge almost surely.

Let the probability space be $[0, 1]$ with the uniform probability measure.

At stage m , partition $[0, 1]$ into m intervals:

$$I_{m,j} = \left[\frac{j-1}{m}, \frac{j}{m} \right), \quad j = 1, \dots, m.$$

List the indicator functions of these intervals sequentially:

$$X_k = \mathbf{1}_{I_{m,j}}$$

for the corresponding pair (m, j) .

At stage m ,

$$\mathbb{P}(X_k = 1) = \frac{1}{m}.$$

Since $m \rightarrow \infty$ as $k \rightarrow \infty$,

$$\mathbb{P}(X_k = 1) \rightarrow 0.$$

Thus, for every $0 < \epsilon < 1$,

$$\mathbb{P}(|X_k| \geq \epsilon) = \mathbb{P}(X_k = 1) \rightarrow 0.$$

Therefore,

$$\boxed{X_k \xrightarrow{P} 0}.$$

However, for every $\omega \in [0, 1]$, at every stage m there is an interval containing ω . Therefore,

$$X_k(\omega) = 1$$

infinitely often.

There are also infinitely many indices for which

$$X_k(\omega) = 0.$$

Hence $X_k(\omega)$ does not converge. Therefore X_k does not converge almost surely.

Thus,

$$\boxed{X_k \xrightarrow{P} 0 \quad \text{but} \quad X_k \not\xrightarrow{\text{a.s.}} 0.}$$

Question 9

(a)

Let X have finite mean

$$\mu = \mathbb{E}[X].$$

Define the nonnegative random variable

$$Y = (X - \mu)^2.$$

By Markov's inequality,

$$\mathbb{P}(Y \geq \alpha^2) \leq \frac{\mathbb{E}[Y]}{\alpha^2}.$$

Since

$$Y \geq \alpha^2 \iff |X - \mu| \geq \alpha$$

and

$$\mathbb{E}[Y] = \text{Var}(X),$$

we obtain

$$\boxed{\mathbb{P}(|X - \mu| \geq \alpha) \leq \frac{\text{Var}(X)}{\alpha^2}.$$

This is Chebyshev's inequality.

(b)

Let

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Since the X_i are IID with mean μ ,

$$\mathbb{E}[S_n] = \mu.$$

By independence,

$$\begin{aligned}\text{Var}(S_n) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{\sigma^2}{n}.\end{aligned}$$

Applying Chebyshev's inequality,

$$\mathbb{P}(|S_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(S_n)}{\epsilon^2}.$$

Therefore,

$$\boxed{\mathbb{P}(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}}.$$

As $n \rightarrow \infty$,

$$\frac{\sigma^2}{\epsilon^2 n} \rightarrow 0.$$

Hence

$$\mathbb{P}(|S_n - \mu| \geq \epsilon) \rightarrow 0.$$

Therefore,

$$\boxed{S_n \xrightarrow{P} \mu}.$$

(c)

We want

$$\mathbb{P}(|S_n - \mu| \leq \epsilon) \geq 1 - \delta.$$

Equivalently, we require

$$\mathbb{P}(|S_n - \mu| \geq \epsilon) \leq \delta.$$

From part (b),

$$\mathbb{P}(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}.$$

Thus it is sufficient that

$$\frac{\sigma^2}{\epsilon^2 n} \leq \delta.$$

Solving for n gives

$$n \geq \frac{\sigma^2}{\epsilon^2 \delta}.$$

Therefore, it is sufficient to choose

$$\boxed{n = \left\lceil \frac{\sigma^2}{\epsilon^2 \delta} \right\rceil}.$$

Then

$$\mathbb{P}(|S_n - \mu| \leq \epsilon) \geq 1 - \delta.$$